

Contrastive Preference Learning

Summary

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1 Problem Setup

Traditional preference-based reinforcement learning uses pairwise comparisons between trajectories to learn reward functions, but this approach can be sample inefficient and computationally expensive. This paper introduces Contrastive Preference Learning (CPL), a method that transforms preference learning into a supervised learning problem using contrastive learning principles. The goal is to learn reward functions more efficiently by leveraging modern supervised learning techniques.

2 Thesis

The central thesis is that preference learning can be reformulated as a contrastive learning problem, allowing for more efficient and scalable reward function learning. The authors argue that by using contrastive objectives and supervised learning techniques, CPL can achieve better performance with fewer computational resources than traditional RL-based preference learning methods.

3 Key Concepts

Contrastive Learning: A machine learning paradigm that learns representations by bringing similar items closer in embedding space while pushing dissimilar items apart.

Preference Segments: Trajectory segments that are compared by humans, with preferences expressed as which segment is better.

Regret Labels: Labels that represent the difference in quality between preferred and non-preferred trajectory segments, derived from optimal value functions.

Supervised Preference Learning: Learning reward functions from preferences using supervised learning techniques rather than reinforcement learning.

Advantage Function: The difference between the Q-value of an action and the value function of a state, representing how much better an action is than average.

4 Methods

The authors develop Contrastive Preference Learning with several key components:

1. **Preference Transformation:** Human preferences are transformed into regret labels using optimal value functions estimated from expert demonstrations.
2. **Contrastive Objective:** A contrastive loss function is used to learn reward functions that assign higher values to preferred segments and lower values to non-preferred segments.

3. **Supervised Learning:** The entire process is framed as supervised learning, eliminating the need for RL-specific algorithms and enabling the use of modern deep learning techniques.
4. **Segment-Based Learning:** The method learns from trajectory segments rather than full trajectories, improving sample efficiency and reducing computational requirements.
5. **Advantage Estimation:** Various methods are explored for estimating the advantage function needed to compute regret labels, with different trade-offs in accuracy and stability.

The key innovation is the transformation of preference learning from an RL problem to a supervised contrastive learning problem.

5 Results

The authors demonstrate CPL’s effectiveness on robotic manipulation tasks:

- **Performance:** CPL significantly outperforms existing preference-based RL methods on MetaWorld benchmarks, achieving higher success rates with less computational cost.
- **Efficiency:** The supervised learning approach is much more computationally efficient than RL-based methods, requiring less training time and computational resources.
- **Sample Efficiency:** CPL achieves better performance with fewer preference queries compared to traditional methods.
- **Scalability:** The method scales well to high-dimensional observation spaces, including image-based inputs.
- **Robustness:** CPL is robust to variations in dataset quality and performs well across different tasks and environments.
- **Ablation Studies:** Systematic experiments show the importance of different components like contrastive objectives and regret labeling.

The results demonstrate that supervised contrastive learning can be highly effective for preference-based reward learning.

6 Significance

This work introduces an important paradigm shift in preference-based reinforcement learning:

- **Paradigm Change:** Transforms preference learning from an RL problem to a supervised learning problem, enabling the use of modern deep learning techniques.
- **Efficiency Gains:** Dramatically improves computational efficiency and sample efficiency compared to traditional RL-based preference learning.
- **Scalability:** Makes preference-based learning more scalable to complex, high-dimensional problems like robotic manipulation with vision inputs.
- **Practical Impact:** The method is more practical for real-world applications where computational resources and human feedback are limited.

- **Research Direction:** Opens up new research avenues for applying supervised learning techniques to traditionally RL-dominated problems.

The paper demonstrates that the benefits of contrastive learning, which have revolutionized computer vision and natural language processing, can also be applied to preference-based reinforcement learning, leading to more efficient and scalable methods.